



Selected Topics on Complex Algebraic Varieties - MPG3320 III
 Assignment 2
 September 28, 2026

Lemma 1: Let $V \subseteq \mathbb{A}_k^n$ be an algebraic variety. Then there exist irreducible affine varieties $V_1, \dots, V_m$ such that

$$V = \bigcup V_i \quad \text{y} \quad V_i \not\subseteq V_j \quad \text{for all } i \neq j$$

Proof. We define

$$\mathcal{C} := \{V \subseteq \mathbb{A}_k^n \text{ affine variety} : V \text{ is not an union of irreducibles}\}$$

If $\mathcal{C}$ is empty there is nothing to prove, otherwise, there exists a minimal $V \in \mathcal{C}$, and by definition of $\mathcal{C}$, it is reducible. This contradicts its minimality.

Let $V = \bigcup V_i$ with V_i irreducible and suppose that $V_i \not\subseteq V_j$ for all $i \neq j$. Suppose that

$$\bigcup V_i = \bigcup W_j \quad \text{where } W_i \not\subseteq W_j \quad \text{and} \quad V_i, W_j \neq \emptyset$$

Notice that $V_1 = V_1 \cap V = \bigcup (V_1 \cap W_j)$, since V_1 is irreducible, there exists a unique j such that $V_1 = V_1 \cap W_j$, that is, $V_1 \subseteq W_j$. In the same way, there exists a unique i such that $W_j \subseteq V_i$, which implies that $V_1 \subseteq V_i$, then $i = 1$ and thus $V_1 = W_j$. $\square$

Since $k[x_1, \dots, x_n]$ is noetherian, every collection of algebraic varieties has a minimal element.

Lemma 2: Let $X \subseteq \mathbb{A}_k^n$ be an algebraic variety and $X = \bigcup X_i$ its decomposition into irreducibles, then

$$\dim(X) = \max \dim(X_i)$$

Proof. Given $V \subseteq X$ an irreducible affine variety, we see that $V = \bigcup (V \cap X_i)$ and since V is irreducible it follows that $V \subseteq X_j$ for some j .

Let $V_0 \subset V_1 \subset \dots \subset V_r$ be irreducible varieties in X , by the previous observation, we have that $V_i \subseteq X_j$ for some j and for all i . Then,

$$r \leq \dim(X_j) \leq \max \dim(X_i) \quad \text{hence} \quad \dim(X) \leq \max \dim(X_i)$$

On the other hand, if $V_0 \subset V_1 \subset \dots \subset V_r$ are irreducible varieties in X_i , in particular, they are so in X , then

$$r \leq \dim(X) \quad \text{and therefore} \quad \dim(X_i) \leq \dim(X)$$

which implies that $\max \dim(X_i) \leq \dim(X)$. $\square$

Both results hold for projective varieties, using identical arguments.

Problem 1. Let us look the following lemma

Lemma 1.1: Let $f : X \rightarrow Y$ be a surjective morphism between irreducible algebraic varieties, then $\dim(Y) \leq \dim(X)$.

Proof. Let us see that the k -algebra morphism $f^* : k[Y] \rightarrow k[X]$ is injective. Let $g \in \ker f^*$, then $(g \circ f)(x) = 0$ for all $x \in X$, in other words,

$$Y = f(X) \subseteq V(g) \quad \text{then} \quad g|_Y \equiv 0$$

Therefore $k(Y) \subseteq k(X)$ and thus $\dim(Y) = \text{trdeg}_k(k(Y)) \leq \text{trdeg}_k(k(X)) = \dim(X)$. $\square$

Let $X = \bigcup X_i$ and $Y = \bigcup Y_j$ be the decomposition into irreducibles of X and Y respectively. Notice that

$$X \times Y = \bigcup X_i \times \bigcup Y_j = \bigcup (X_i \times Y_j)$$

and since the product of irreducible varieties is irreducible, it follows that $\bigcup (X_i \times Y_j)$ is the decomposition into irreducibles of $X \times Y$. Let X_i and Y_j of maximal dimension, by Noether's normalization lemma there exist surjective morphisms

$$f : X_i \rightarrow \mathbb{A}_k^{\dim(X_i)} \quad \text{and} \quad g : Y_j \rightarrow \mathbb{A}_k^{\dim(Y_j)}$$

We have a surjective morphism between irreducible varieties $\varphi = (f, g) : X_i \times Y_j \rightarrow \mathbb{A}_k^{\dim(X_i) + \dim(Y_j)}$, then

$$\dim(X) + \dim(Y) = \dim(X_i) + \dim(Y_j) \leq \dim(X \times Y)$$

On the other hand, let $X_i \times Y_j$ be of maximal dimension. By Noether's normalization lemma, $k[X_i]$ is algebraically generated by $\dim(X_i)$ elements, and similarly, $k[Y_j]$ is algebraically generated by $\dim(Y_j)$ elements.

Therefore, $k[X_i \times Y_j] \cong k[X_i] \otimes_k k[Y_j]$ is algebraically generated by $\dim(X_i) + \dim(Y_j)$ elements. Thus, we see that

$$\begin{aligned} \dim(X \times Y) &= \dim(X_i \times Y_j) = \text{trdeg}_k(k[X_i \times Y_j]) \\ &= \text{trdeg}_k(\text{Frac}(k[X_i] \otimes_k k[Y_j])) \leq \dim(X_i) + \dim(Y_j) \leq \dim(X) + \dim(Y) \end{aligned}$$

Which finalizes the proof.

Problem 2. Let us see φ is surjective. Let $[a] \in \mathbb{P}_k^n$, we consider the hyperplane $V^p(h_a) \in (\mathbb{P}_k^n)^\vee$, where $h_a = a_0x_0 + \dots + a_nx_n$, note that $\varphi(V^p(h_a)) = [a]$. Let $[a] = [b]$, then there exists $\lambda \neq 0$ such that $a = \lambda b$, hence

$$h_a = \sum_{i=0}^n a_i x_i = \sum_{i=0}^n \lambda b_i x_i = \lambda \sum_{i=0}^n b_i x_i = \lambda h_b$$

so h_a and h_b differ by a non-zero scalar, therefore they define the same hyperplane, in other words $V^p(h_a) = V^p(h_b)$. We define

$$X = \{V^p(h_a) : [1 : 0 : \dots : 0] \in V^p(h_a)\}$$

and $V = \varphi(X)$. We claim that $V = V^p(a_0)$. Indeed, if $[a] \in V^p(a_0)$ then $a_0 = 0$ and therefore

$$h_a = \sum_{i=1}^n a_i x_i \quad \text{then} \quad h_a(1, 0, \dots, 0) = 0$$

and thus $[a] \in V$. On the other hand, if $V^p(h_a) \in X$ we have that $0 = h_a(1, 0, \dots, 0) = a_0$, then $[a] \in V^p(a_0)$. Which concludes what was requested.

Problem 3.

- (a) Since the ideal $I = (y^2 - x^3 - ax - b)$ is generated by a single element, its homogenization is equal to the ideal $(y^2z - x^3 - axz^2 - bz^3)$, then

$$\overline{V(I)} = V^p(\tilde{I}) = V^p(y^2z - x^3 - axz^2 - bz^3)$$

Then, if $z = 0$ we have that $x = 0$ and therefore $y = 1$. We conclude that the projective closure of $V(I)$ added a unique point at infinity, which corresponds to $[0 : 0 : 1]$.

- (b) We want to show that $\{x_1^2 - x_2^2 - 1, x_3 - x_1\}$ is a Gröbner basis for $I = (x_1^2 - x_2^2 - 1, x_3 - x_1)$ with the lexicographic monomial order given by $x_3 \geq x_1 \geq x_2$. Under this order,

$$LM_{\leq}(x_1^2 - x_2^2 - 1) = x_1^2 \quad \text{y} \quad LM_{\leq}(x_3 - x_1) = x_3$$

It suffices to show that, given $f \in I$, its leading monomial is divisible by x_3 or x_2^2 . Since $f \in I$, there exist $g, h \in k[x_1, x_2, x_3]$ such that

$$f = g \cdot (x_1^2 - x_2^2 - 1) + h \cdot (x_3 - x_1)$$

We have two cases: if the leading monomial of f has at least one factor x_3 , then it is divisible by x_3 . If it does not, then no monomial contains it as a factor; otherwise it would be the leading monomial. Thus, x_3 is a free variable, so

$$f(x_1, x_2, x_3) = f(x_1, x_2, x_1) = g(x_1, x_2, x_1) \cdot (x_1^2 - x_2^2 - 1)$$

Then, the leading monomial of f is the leading monomial of g multiplied by x_1^2 , since the order respects multiplication. Therefore, $\overline{W} = V^p(x_1^2 - x_2^2 - x_0^2, x_3 - x_1)$. If $x_0 = 0$, we have

$$x_1^2 = x_2^2 \quad \text{then} \quad x_1 = \pm x_2$$

Since $x_3 = x_1$, it follows that $x_1 \neq 0$. Therefore, the points at infinity are

$$[0 : 1 : 1 : 1] \quad \text{and} \quad [0 : 1 : -1 : 1]$$

Problem 4. The twisted cubic is given by the ideal $(y - x^2, z - x^3)$, that is, $Y = V(y - x^2, z - x^3)$ and it is parameterized by the expression (t, t^2, t^3) . We see that

$$k[x, y, z]/(y - x^2, z - x^3) \cong k[x]$$

and therefore $(y - x^2, z - x^3)$ is a prime ideal, so by the Nullstellensatz we have that

$$I(Y) = \sqrt{(y - x^2, z - x^3)} = (y - x^2, z - x^3) =: I$$

Let $J = (yw - x^2, xz - y^2, zw - xy)$ and note that $J \subseteq \tilde{I}$, since the generators of J are in $\tilde{I}$. It follows that $\overline{Y} = V^p(\tilde{I}) \subseteq V^p(J)$, we would like to prove that $V^p(J) = \overline{Y}$. Consider the morphism

$$\begin{aligned} \phi : \mathbb{P}_k^1 &\rightarrow V^p(J) \\ [s : t] &\mapsto [s^3 : s^2t : st^2 : t^3] \end{aligned}$$

Evaluating at the generators of J we see that ϕ is well-defined, moreover, its inverse is given by

$$[w : x : y : z] \mapsto \begin{cases} [w : x] & \text{si } w \neq 0 \text{ ó } x \neq 0 \\ [y : z] & \text{si } y \neq 0 \text{ ó } z \neq 0 \end{cases}$$

Therefore, ϕ is a homeomorphism, in particular, it is a closed map. Since $\phi(U_0) \subseteq Y$, we have the following

$$V^p(J) = \phi(\mathbb{P}_k^1) = \phi(\overline{U_0}) = \overline{\phi(U_0)} \subseteq \overline{Y}$$

Suppose, by contradiction, that $\tilde{I} = (yw - x^2, zw^2 - x^3)$, then $V^p(yw - x^2, zw^2 - x^3) = \overline{Y} = V^p(J)$. The parameterization ϕ tell us that there is a unique point Y such that $w = 0$, namely, $[0 : 0 : 0 : 1]$. However, in the variety $V^p(yw - x^2, zw^2 - x^3)$ we have a copy of $\mathbb{P}_k^1$, since $x = 0$ and y, z remain free. This contradiction proves the claim.

Problem 5. No answer.

Problem 6.

- (a) Clearly the projection is a morphism of algebraic varieties, let us see that it is finite and surjective. In the first place, note that $k[\mathbb{A}_k^2] = k[x, y]$, then the induced morphism

$$\pi^* : k[x, y] \rightarrow k[X]$$

is such that $x \mapsto x$ and $y \mapsto y$. Furthermore, given $p \in k[x, y, z]$ we have that $\deg_z(p(z^3 + xz - y)) \geq 1$, which implies that π^* is injective. Moreover, the element z satisfies the equation

$$z^3 + xz - y = 0$$

so $k[X]$ is generated by $\{1, z, z^2\}$ as a $k[x, y]$ -module, that is, it is finitely generated. Therefore, $k[X]$ is an integral extension of $k[x, y]$. This proves that ϕ is a finite, surjective morphism.

(c) Since k is a field of characteristic 0, we have $\mathbb{Q} \subseteq k$. Let $(a, b) \in \mathbb{A}_k^2$, then

$$\pi^{-1}(a, b) = \{(a, b, z) \in \mathbb{A}_k^3 : z^3 + az - b\}$$

Then z corresponds to the root of a cubic equation, whose solutions are given by the Cardano-Tartaglia formula, namely

$$z_1 = \sqrt[3]{\frac{b}{2} + \sqrt{\frac{b^2}{4} + \frac{a^3}{27}}} + \sqrt[3]{\frac{b}{2} - \sqrt{\frac{b^2}{4} + \frac{a^3}{27}}} = u + v$$

The other two solutions are $z_2 = \omega u + \omega^2 v$ and $z_3 = \omega^2 u + \omega v$, where ω is a cube root of unity distinct from 1. If (a, b) are such that

$$\frac{b^2}{4} + \frac{a^3}{27} \neq 0$$

so $\pi^{-1}(a, b)$ consists of three distinct elements. Indeed, if $z_1 = z_2$ then we have

$$0 = z_1 - z_2 = (1 - \omega)u + (1 - \omega^2)v$$

which implies that

$$\begin{aligned} 0 &= (1 - \omega)^3 u^3 + (1 - \omega^2)^3 v^3 = -3\omega(1 - \omega)u^3 - 3\omega^2(1 - \omega^2)v^3 \\ &= -3\omega(1 - \omega)u^3 + 3\omega(1 - \omega)v^3 = 3\omega(1 - \omega)(v^3 - u^3) \end{aligned}$$

Therefore, $u^3 = v^3$, it follows that $\frac{b^2}{4} + \frac{a^3}{27} = 0$, this prove that $z_1 \neq z_2$. The other cases are obtained similarly. Thus, π is a covering map of degree 3 outside of

$$B = V\left(\frac{b^2}{4} + \frac{a^3}{27}\right)$$

Then, the branch locus of π corresponds to B .

(b) Since $\mathbb{Q} \subseteq k$, we can write

$$\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \quad \text{and} \quad \omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i \quad \text{where } i \text{ is a root of } -1$$

In B we have two options, $(a, b) = 0$ ó $(a, b) \neq 0$ in which case $a \neq 0$ and $b \neq 0$. In the first one we have

$$\pi^{-1}(0, 0) = (0, 0, 0)$$

and in the second case, replacing the value of ω and ω^2 into z_2 and z_3 , we obtain the following

$$\pi^{-1}(a, b) = \left\{ \left(a, b, 2\sqrt[3]{b/2} \right), \left(a, b, -\sqrt[3]{b/2} \right) \right\}$$

Thus,

$$R = \pi^{-1}(B) = \{0\} \cup \left\{ \left(a, b, 2\sqrt[3]{b/2} \right), \left(a, b, -\sqrt[3]{b/2} \right) : (a, b) \in B \right\}$$

(d) Outside of B , the fiber of a point has 3 distinct elements and each with multiplicity 1. In B we have two cases, the origin, in which case the fiber consists of a unique point with multiplicity 3. And in the second case, we have two distinct points in the fiber, one with multiplicity 1 (z_1) and another with multiplicity 2 ($z_2 = z_3$).

Problem 7. The segre embedding of $\mathbb{P}_k^1 \times \mathbb{P}_k^1$ into $\mathbb{P}_k^3$ is given by the map

$$\begin{aligned} \sigma_{1,1} : \mathbb{P}_k^1 \times \mathbb{P}_k^1 &\rightarrow \mathbb{P}_k^3 \\ ([x : y], [a : b]) &\mapsto [xa : xb : ya : yb] \end{aligned}$$

Whose image corresponds to the projective variety given by the equation $XW - YZ$. Note that the expression is equivalent to the zero locus of a quadratic form, indeed,

$$XW - YZ = [X : Y : Z : W] \begin{pmatrix} 0 & 0 & 0 & 1/2 \\ 0 & 0 & -1/2 & 0 \\ 0 & -1/2 & 0 & 0 \\ 1/2 & 0 & 0 & 0 \end{pmatrix} [X : Y : Z : W]^T$$

Let B be the matrix associated with such quadratic form, we see that $\text{rank}(B) = 4$. As the matrix A , which defines the quadratic form Q , has rank 4, by linear algebra there exists P invertible such that $P^T A P = B$. We define

$$\begin{aligned}\varphi : \mathbb{P}_k^3 &\rightarrow \mathbb{P}_k^3 \\ [x] &\mapsto [P^{-1}x]\end{aligned}$$

Let us see that $\varphi(X) = \Sigma_{1,1}$. Let $[x] \in X$, then

$$\varphi(x)^T B \varphi(x) = (P^{-1}x)^T P^T A P (P^{-1}x) = x^T (P^{-1})^T P^T A P (P^{-1}x) = x^T A x = 0$$

and therefore $\varphi([x]) \in \Sigma_{1,1}$. On the other hand, let $[y] \in \Sigma_{1,1}$ and we consider $[x] = [Py]$, we have

$$x^T A x = (Py)^T A P y = y^T P^T A P y = y^T B y = 0$$

that is, $[y] \in \varphi(X)$. Which concludes the proof.